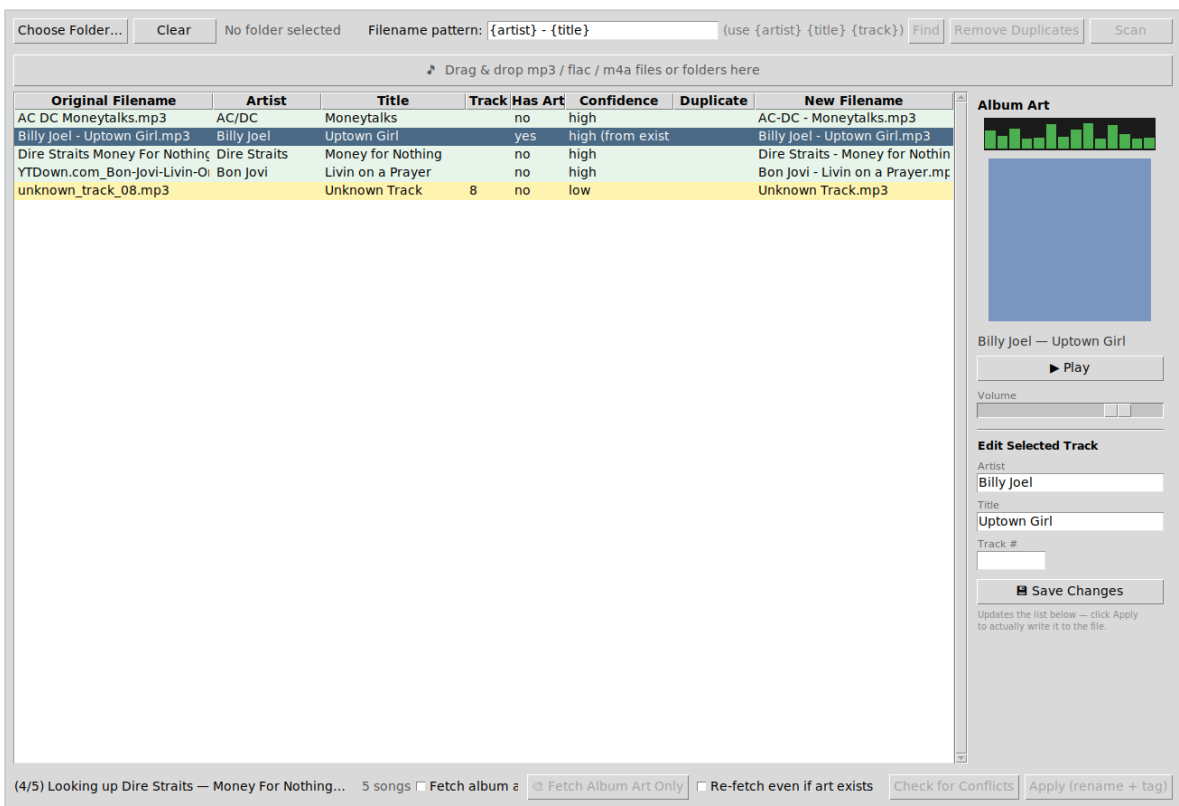


Music Standardizer

User Manual

A desktop tool that cleans up messy music-library filenames and tags — fixes artist / title / track metadata, embeds album art, finds duplicates, and lets you review every change before anything is touched.



The main window: proposed changes previewed in the table, album art and playback on the right.

Version covered: single-file GUI + CLI build (music_standardizer.py)

Works with: MP3, FLAC, M4A / MP4 audio files

Contents

1. Getting Started
2. Installing (Windows / Mac / Linux)
3. The Main Window, at a Glance
4. Step-by-Step Workflow
5. Understanding the Table & Confidence Colors
6. Album Art
7. Looking Up Unclear Tracks Online
8. Manual Corrections
9. Finding & Removing Duplicates
10. Checking for Filename Conflicts
11. Playback & the Side Panel
12. Customizing Artist Names (the Whitelist)
13. Command-Line Usage (Advanced)
14. Building a Standalone .exe
15. Troubleshooting
16. Good Habits & Safety Notes

1. Getting Started

Music Standardizer takes a folder full of messy filenames — the kind left behind by YouTube downloaders and random rippers — and turns them into clean, correctly tagged files, with cover art, while showing you exactly what will change before you commit to anything.

A typical before-and-after:

```
YTDn.com_YouTube_AC-DC-Moneytalks-Official-HD-Video_Media_2lqdErI9uss_009_128k.mp3
↓
AC-DC - Moneytalks.mp3 (tagged: Artist = AC/DC, Title = Moneytalks, art embedded)
```

Nothing is renamed, retagged, or moved until you click **Apply**. Every step before that is just a preview.

2. Installing

Option A — Run the Python script (recommended for most people)

Requires Python 3.10 or later.

```
pip install mutagen tkinterdnd2 pygame-ce pillow
python music_standardizer.py
```

What each package enables — the app still runs without any of them, it just quietly disables that one feature and tells you why in the window:

Package	Enables
mutagen	Reading and writing audio tags (required for everything — always install this one)
tkinterdnd2	Drag-and-drop of files/folders onto the window
pygame-ce	In-app playback (play button + volume slider)
pillow	Album art preview inside the app

Windows + Python 3.13/3.14 users

The original pygame package fails to build on very new Python versions (it needs a compiler module Python removed). Use the community fork pygame-ce instead — same `import pygame` code, prebuilt wheels, no build step. Don't install both packages at once; they share the same import name and will conflict.

Option B — Use a pre-built .exe (Windows, no Python needed)

If someone has already built the standalone executable for you (see Section 14), just double-click it. No installation step is required, but drag-and-drop and playback only work if the exe was built with those features included.

3. The Main Window, at a Glance

The window is split into three areas:

- **Top bar** — Choose Folder, Clear, the filename pattern, Scan, and the duplicate-finder buttons.
- **Center table** — one row per audio file: current filename, guessed Artist / Title / Track, whether it has art, how confident the guess is, duplicate status, and the proposed new filename. Click any column header to sort by it.
- **Right side panel** — album art preview, a small visualizer, playback controls with volume, and fields to manually correct the selected track.

The bottom bar shows a live status message, a running count of how many songs are listed, and the action buttons: **Apply**, **Check for Conflicts**, and **Fetch Album Art Only**, plus checkboxes for Force Art, Fetch Art, and Look up unclear tracks online.

4. Step-by-Step Workflow

- 1 Click **Choose Folder...** and pick your music folder — subfolders are scanned automatically. You can also just drag files or folders straight onto the window (if drag-and-drop is available).
- 2 The table fills in with every audio file found, each with a guessed Artist / Title / Track and a proposed new filename. Nothing has been touched yet.
- 3 Look for yellow rows — these are **low-confidence** guesses worth a glance. Double-click any Artist / Title / Track cell to fix it directly in the table, or select the row and use the Edit panel on the right (Section 8).
- 4 Turn on **Fetch album art** and/or **Look up unclear tracks online** if you want the app to pull cover art and correct uncertain guesses from iTunes' free catalog (Sections 6–7).
- 5 Click **Check for Conflicts** to catch any filename collisions before they happen (Section 10) — optional but recommended for large batches.
- 6 Click **Apply**. This is the only step that renames files and writes tags. The status bar shows live progress, and a summary appears when it's done.

Nothing happens until you click Apply

Scanning, previewing, sorting, and editing are all read-only. Your files are only renamed or retagged when you press the Apply button.

5. Understanding the Table & Confidence Colors

Column	Meaning
Filename	The file's current name on disk.
Artist / Title / Track	The app's best guess, editable by double-clicking.
Has Art	Whether the file already has embedded cover art.
Confidence	How sure the app is about Artist/Title — see color key below.
Duplicate	Blank, "keep", or "remove" — set by Find Duplicates (Section 9).
Proposed Filename	What the file will be renamed to on Apply.

Row Color	Meaning
Yellow	Low confidence — the filename didn't clearly split into artist/title. Worth checking.
Green	High confidence — parsed cleanly, or confirmed from the file's existing tags.
Red	Marked as a duplicate to be removed (after running Find Duplicates).

Click any column header once to sort by it, ascending; click again to reverse the order. An arrow (▲ / ▼) on the header shows the active sort.

6. Album Art

Turn on the **Fetch album art** checkbox before clicking Apply to have the app search iTunes' free public catalog for cover art and embed it directly into each file's tags. Files that already have art are left alone unless you also check **Force Art**, which overwrites existing art with the new match.

If you only want art — without renaming files or changing any other tag — use the dedicated **Fetch Album Art Only** button instead. It runs the same lookup and embedding, but never touches the filename, artist, or title.

Rate limiting is handled automatically

iTunes' free search API allows only a limited number of lookups per minute. For a large library, the app paces its own requests to stay under that limit — if it needs to pause, the status bar will say so ("rate limit reached — waiting...") instead of the window appearing frozen. You don't need to do anything; just let it run.

7. Looking Up Unclear Tracks Online

Turn on **Look up unclear tracks online** to have any low-confidence (yellow) row's raw filename searched against iTunes' catalog, so the real artist and title can replace a bad guess before the file is renamed. This is a best-effort correction, not a guarantee — a track with no match, or an unusual live/cover/remix version, may still come back unresolved, and you can always fix it by hand (Section 8).

When a lookup also returns cover art and **Fetch album art** is on, the app reuses that same result instead of making a second network request for the same track.

8. Manual Corrections

Two ways to fix a wrong guess:

- **Quick fix:** double-click the Artist, Title, or Track cell directly in the table and type the correct value.
- **Full edit:** click the row to select it, use the Artist / Title / Track fields in the "Edit Selected Track" section of the side panel, and click **Save Changes**.

Either way, the row's confidence updates to "high (manual)" and the Proposed Filename column recalculates immediately — but again, nothing is written to disk until Apply.

9. Finding & Removing Duplicates

Click **Find Duplicates** to scan the currently listed files for tracks that share the same Artist and Title. Within each duplicate group, the app keeps the best copy — preferring one that already has embedded art, then one with confidently-known tags, then the larger file (a rough proxy for a better-quality rip) — and marks the rest "remove" (shown in red).

Click **Remove Duplicates** to act on those marks. Files are **moved, not deleted** — each one goes into a "Duplicates" subfolder next to the original, so the action is fully reversible if you change your mind.

Matching is by Artist + Title, not filename

Two files are treated as duplicates only if their parsed (or tagged) Artist and Title match exactly. Two copies of the same song with clean, matching tags will always be caught — even if their filenames look nothing alike. The one gap: if a duplicate's filename carries an extra suffix that changes the parsed title, such as "Song (1).mp3" or "Song copy.mp3", and neither file already has a correct embedded title tag, that pair may be parsed as two different titles and won't be grouped. Fix the title manually first (Section 8) if that happens, then re-run Find Duplicates.

10. Checking for Filename Conflicts

Click **Check for Conflicts** at any time before Apply to see whether any proposed new filenames would collide — either with each other (two files in this batch that would end up with the same name) or with a file that already exists on disk. The app reports every conflict it finds so you can fix the underlying

Artist/Title/Track values before committing. This step makes no changes on its own; it's purely informational.

11. Playback & the Side Panel

Click any row to preview it: the side panel shows its album art (if available), a small decorative visualizer above it, and a **Play** button with a volume slider. Playback requires the `pygame-ce` package (Section 2) — without it, the button is disabled and the panel explains why.

12. Customizing Artist Names (the Whitelist)

Names with unusual punctuation — AC/DC, Jay-Z, T-Pain — get mangled by naive filename splitting, so the app ships with a built-in whitelist of common tricky names and matches them regardless of whether the original filename used spaces, hyphens, or underscores ("AC DC", "AC-DC", and "AC_DC" all resolve the same way).

To teach the app a name specific to your library, open `music_standardizer.py` in a text editor and add an entry to the `ARTIST_WHITELIST` dictionary near the top of the file:

```
ARTIST_WHITELIST = {  
    "ac-dc": "AC/DC",  
    "my-obscure-band": "My Obscure Band",  
    # add your own entries here  
}
```

For the command-line tool, you can instead pass a whitelist file with `--whitelist yourfile.txt`, one entry per line, formatted `raw-as-it-appears|Canonical Name`.

13. Command-Line Usage (Advanced)

For scripting or headless use, run the same file with `scan` or `apply` as the first argument instead of launching the GUI.

Step 1 — Scan a folder and write a review sheet

```
python music_standardizer.py scan "/path/to/Music" -o review.csv
```

Open `review.csv` in any spreadsheet program and fix any wrong guesses in the artist / title / track columns.

Step 2 — Apply the (edited) CSV

```
python music_standardizer.py apply review.csv
```

Preview without touching anything

```
python music_standardizer.py apply review.csv --dry-run
```

Also fetch album art and look up unclear rows online

```
python music_standardizer.py apply review.csv --art --online-lookup
```

Flag	Effect
--dry-run	Shows what would happen; makes no changes.
--art	Fetches and embeds cover art from iTunes.
--force-art	Also overwrites files that already have art (implies --art).
--online-lookup	Corrects low-confidence rows using iTunes' text search.
--whitelist FILE	Loads extra artist-name corrections from a text file.

14. Building a Standalone .exe

To share the app with someone who doesn't have Python installed, build a single-file executable with PyInstaller. Run this on the same operating system you want the executable to run on (a Windows .exe must be built on Windows).

```
pip install pyinstaller mutagen tkinterdnd2 pygame-ce pillow

pyinstaller --onefile --windowed --name MusicStandardizer \
--collect-all tkinterdnd2 \
--hidden-import PIL._tkinter_finder \
music_standardizer.py
```

The finished executable appears in the newly created dist folder. The two extra flags matter: without `--collect-all tkinterdnd2`, drag-and-drop will silently fail in the built exe even though it works fine when run from source; without `--hidden-import PIL._tkinter_finder`, the album art preview can fail the same way.

15. Troubleshooting

Symptom	Likely cause & fix
Drag-and-drop shows a "no-drop" cursor, or the window says drag-and-drop is unavailable	tkinterdnd2 isn't installed, or the exe was built without --collect-all tkinterdnd2. Reinstall/rebuild with that flag.
pip install pygame fails on Windows with a distutils error	You're on Python 3.13/3.14, which removed the module pygame's old build script needs. Install pygame-ce instead (Section 2).
Album art preview doesn't show in the built exe, but works from source	Rebuild with --hidden-import PIL._tkinter_finder (Section 14).
Play button is greyed out	pygame-ce isn't installed, or the file format isn't supported for playback.
Album art / online lookup seems to pause partway through a big batch	This is the built-in rate limiter pacing requests to iTunes' free API — normal for libraries of 50+ tracks. The status bar explains the wait; just let it finish.
A duplicate pair wasn't detected	Their parsed titles differ, often because of a suffix like "(1)" in one filename. See the tip in Section 9.

16. Good Habits & Safety Notes

- **Always review the table before clicking Apply**, especially yellow (low-confidence) rows.
- **Removed duplicates are moved, not deleted** — check the "Duplicates" subfolder before permanently deleting anything yourself.
- Renaming only happens if the destination filename doesn't already exist — the app will report a conflict rather than overwrite a file silently.
- Consider testing on a copy of a small folder first if you're working with an irreplaceable collection, until you're comfortable with how the tool behaves.

End of manual — for source code and customization, see `music_standardizer.py`.